



Rohan Dhanraj Yadav

AI/ML ENGINEER · GENERATIVE AI · AGENTIC SYSTEMS ·
AI PLATFORM DEVELOPER

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7+

TOTAL EXPERIENCE

4+

AI/ML EXP

1.5y

AGENTIC AI

3y

GEN AI

8+

AI PROJECTS

15

TEAM SIZE

6+

F500 CLIENTS

PROFESSIONAL PROFILE

Results-driven AI/ML Engineer with **7+ years of professional experience** including **4+ years specializing in designing, developing, and deploying production-grade AI systems** across enterprise domains. Deep expertise in **Generative AI, Agentic AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration, and Machine Learning**.

Proven track record in leading cross-functional teams of **15+ engineers**, architecting scalable ML/AI pipelines, and delivering high-impact AI applications for **Fortune 500 clients** including Cisco, ITC Fairview, KDP, Omnitrax, PDS, and Marco Technologies. Known for building production systems handling **4,500+ daily API requests**, processing **100K+ daily records**, and achieving **18-38% model performance improvements** in production environments.

TECHNICAL SKILLS

PROGRAMMING LANGUAGES & FRAMEWORKS

Python, Django, Flask,
FastAPI, Streamlit

GENERATIVE AI & LLM ENGINEERING

LLMs: OpenAI, Anthropic,
Vertex AI, Google Gen AI
Platform, Cohere, AI21 Labs
Frameworks: LangChain,
LangGraph
Techniques: Prompt
Engineering, Structured
Outputs, RAG, Prompt
Chaining, Chain-of-Thought,
Graph-Based Gen, LLM
System Design/Architecture

AGENTIC AI & MULTI- AGENT SYSTEMS

Design Patterns: Reflexion,
ReAct, Self-Ask, Plan &
Execute, Tool Use, Human-
in-the-loop, Tree of Thought,
Multi-Agent Collaboration,
Deep Research
Techniques: Agent
Orchestration, LangGraph
Workflows, Multi-Agent
Systems, Tool Integration,
Memory Management,
Autonomous Decision-
Making & Agents

MACHINE LEARNING & DEEP LEARNING

Scikit-Learn, TensorFlow, Keras, PyTorch, Optuna, HuggingFace Transformers, Model Fine-tuning, Hyperparameter Optimization, PyCaret

NATURAL LANGUAGE PROCESSING (NLP)

Spacy, NLTK, Transformers (Encoder/Decoder/Encoder-Decoder), Sentiment Analysis, NER, Text Classification, Zero/One-Shot, Aspect-Based Sentiment, Text Generation, Summarization, Q&A, Translation, Paraphrasing

DATA ENGINEERING & DATABASES

Apache Airflow, ETL, SQL (MySQL, PostgreSQL, BigQuery), NoSQL (MongoDB, Cassandra), Vector DBs (Pinecone, Qdrant, LanceDB, Chroma, Pgvector, Milvus, Weaviate)

ANALYTICS & STATISTICAL COMPUTING

NumPy, Pandas, Matplotlib, Seaborn, Plotly, SciPy, Statistical Analysis, Predictive Analytics, Data Visualization

WEB SCRAPING & DATA EXTRACTION

BeautifulSoup, Scrapy, Selenium, Playwright, Pydantic Structured Output

DEVOPS, CLOUD & MLOPS

Git, GitHub, Docker, CI/CD (GitHub Workflows, Jenkins), Google Cloud Platform (GCP), Amazon Web Services (AWS), MLOps

WORK EXPERIENCE

AI Engineer — Backend & AI Systems Architecture

November 2024 – October 2025

[AI Tech Solutions Ltd.](#) | Full-time

Focus: Architecting production-ready Agentic AI platforms, advanced RAG systems, and multi-agent orchestration

- Led 15-member team building scalable AI systems using Python, FastAPI for backend services and production deployment
- Developed modular ML pipelines (Omodore platform) achieving **38% accuracy improvement** through advanced RAG architecture
- Designed multi-agent orchestration systems using **LangGraph** for complex workflow automation and autonomous decision-making
- Implemented prompt chaining and **chain-of-thought reasoning** mechanisms
- Deployed containerized AI systems using **Docker** with vector databases (Qdrant, LanceDB, Chroma)
- Built automated web scraping systems (Rbetraj) with **Playwright** and **Selenium** for real-time data extraction
- Architected multi-modal virtual agent (GALAMBO) integrating GPT-4 and autonomous tool use

Python

LangChain

LangGraph

FastAPI

LLMs

RAG

Docker

Vector DBs

Software Engineer — Generative AI | ML | Data Science | Python

September 2022 – October 2024

[Aimlytics Technology](#) | Full-time

Focus: Enterprise-scale Generative AI, NLP, and ML solutions for Fortune 500 clients

- Led team of 8 engineers delivering AI solutions for **Fortune 500 clients**
- Built RAG-based virtual sales assistant (SalesMojito) handling **4,500+ daily API requests**
- Built ML pipelines processing **100K+ daily records** using **Apache Airflow**
- Fine-tuned **BERT** and **RoBERTa** models achieving **18% performance improvement**
- Implemented vector database solutions (Qdrant, Chroma) for semantic search
- Built multi-modal educational AI system (InventtEd)
- Developed multilingual NLP pipeline supporting **15+ languages** (evalmyBRAND)
- Resolved **28 critical production issues** with 100% on-time delivery

Airflow

BERT

RoBERTa

FastAPI

GCP

Qdrant

Machine Learning Engineer Intern

August 2021 – September 2022

[iNeuron.ai](#) | [Internship](#)

Focus: ML pipeline development and optimization

- Built ML pipeline architectures for multi-class classification
- Reduced training time by **35%** using **Optuna**
- Deployed ML applications using Flask and Docker
- Built protein classification system (77 markers, 8 classes)

Scikit-Learn

Optuna

Flask

Docker

AWS

Metallurgical Quality Engineer

August 2016 – November 2019

[Bureau Veritas Group](#) | [Full-time](#)

Focus: Quality assurance and industrial data analytics

- Conduct metallurgical testing and quality assurance across major steel companies
- Apply Python (NumPy, Pandas, SciPy, Plotly) for statistical analysis and data-driven quality improvement
- Manage quality compliance and predictive analysis for multiple production facilities

PROJECT SYNOPSIS

Project 1: OMODORE

Nov 2024 - Oct 2025

Domain: AI Agent Platform | Role: Senior AI Engineer

Omodore is an auto-agent platform enabling non-technical users to build domain-specific AI agents without coding expertise. The platform leverages advanced RAG pipelines to deliver accurate, context-aware responses with a 45% accuracy boost.

- Led 15-member team developing Gen AI and Agentic AI solutions
- Architected scalable RAG pipelines using LangChain and LangGraph
- Implemented prompt chaining and Chain of Thoughts methodology for improved reasoning

- Built automated agent building workflows using FastAPI
- Integrated multiple LLMs for optimal performance and Qdrant/Milvus for retrieval

Python

FastAPI

LangGraph

RAG

Docker

Project 2: RBETRAJ

Nov 2024 - Oct 2025

Domain: E-commerce Data Intelligence | Role: Senior AI Engineer

Rbetraj is an Automated Dropshipping Platform powered by Shop Scan Agent specializing in e-commerce data extraction with advanced price tracking. Includes open-source scraper AI TechCrawler for extraction and insights.

- Developed open-source automated scraping framework AI Tech Crawler
- Implemented GPT-powered and LLM-based structured data extraction with Pydantic
- Created robust scraping pipelines using Playwright and BeautifulSoup
- Designed RESTful APIs using FastAPI for distributed data access

Playwright

FastAPI

Pydantic

BeautifulSoup

Docker

Project 3: GALAMBO

Nov 2024 - Oct 2025

Domain: Multi-Modal AI Assistant | Role: Senior AI Engineer

Sophisticated multi-modal virtual agent combining context-aware GPT responses with advanced image search and autonomous tool use.

- Architected multi-modal AI system integrating text and image processing
- Implemented autonomous tool use and search engine integration for information retrieval
- Built scalable API infrastructure using FastAPI and optimized model response times

Multi-modal AI

GPT-4

LangChain

FastAPI

Project 4: SALESMOJI

Nov 2023 - Oct 2024

Domain: Sales and Marketing | Role: Software Engineer (Gen AI | ML)

Harnesses GPT and LLMs to answer sales questions in plain English. Processes 1M+ daily records through Airflow pipelines.

- Led 8-member team delivering custom AI workflows for Fortune 500 clients
- Built RAG-based Virtual Sales Assistant handled 10K+ daily queries with 89% accuracy
- Architected ML pipelines for predictive sales forecasting and trend analysis
- Implemented vector DB solutions (Qdrant, Milvus, Weaviate) for efficient retrieval

GPT

RAG

Airflow

Milvus

TensorFlow

Project 5: AISERA

July 2023 - Oct 2024

Domain: Customer Service and IT Operations | Role: Software Engineer (AI | ML)

Enterprise Generative AI platform for automating customer support through NLP and ML workflows.

- Led team of 8 (5 Data Eng + 3 Soft Dev) delivering workflows for Cisco, ITC, etc.
- Fine-tuned BERT models achieving 50% performance improvement in sentiment analysis
- Implemented custom field prediction using advanced BERT architectures and Airflow triggers
- Architected scalable ML pipelines and integrated LangChain for conversational AI

BERT

LangChain

Chroma

FastAPI

GCP

Project 6: EVALMYBRAND

Sep 2022 - Oct 2024

Domain: Customer Experience | Role: Software Developer (AI | ML)

AI-powered platform supporting 15+ languages to turn unstructured feedback into brand insights and sentiment tracking.

- Fine-tuned RoBERTa for Aspect-Based Sentiment Analysis with 50% performance gain
- Implemented BERT for Neural Machine Translation with 27% accuracy improvement
- Built innovative pipeline for mixed-code translation using Google/LibreOffice APIs
- Orchestrated large-scale data pipelines using Apache Airflow and Oracle/Cassandra

RoBERTa

NMT

Airflow

FastAPI

Poetry

Project 7: INVENTED

Jan 2023 - Aug 2023

Domain: EdTech | Role: Software Developer (AI | ML)

Pioneering EdTech platform providing AI-powered tools for summarization, essay grading, and performance prediction.

- Fine-tuned CLIP for text-to-image generation and Flan-T5 for keyword-based MCQ generation
- Designed scaled Phishing Classifier and multi-modal social media classification model
- Integrated multiple AI models into a cohesive platform with optimized Transformers
- Performed advanced prompt engineering for ChatGPT-based virtual tutoring

CLIP

Flan-T5

Prompt Eng

GCP

Project 8: MICE PROTEIN EXPRESSION

Aug 2021 - Sep 2022

*Domain: Bioinformatics | Role: ML Engineer Intern**Multi-class classification of 77 protein markers in Down syndrome research involving complex biological datasets.*

- Led team of 10 to architect end-to-end ML system with automated pipelines
- Achieved 40% training time reduction using Optuna hyperparameter tuning
- Developed large-scale biological data preprocessing and Flask-based model serving
- Integrated MySQL and Cassandra DB for efficient data management

Optuna

Flask

Cassandra

PyCaret

AWS

EDUCATION

B.Tech. Mechanical Engineering

2016

[Biju Patnaik University of Technology, Odisha](#)*GATE Qualified — Credential ID: A2WR3*

CERTIFICATIONS



GATE ME 2016

Credential ID: A2WR3



Machine Learning Zoom Camp

DataTalksClub | Credential ID: 8F163B



Gold Badge: Python

HackerRank | Credential ID: 6920934A49FA



Gold Badge: Problem Solving

HackerRank | Credential ID: 113775E95540



10 Days of Statistics

Gold Badge | HackerRank



30 Days of ML Challenge

Kaggle | Rank 69 / 7,573 teams



ACHIEVEMENTS & AWARDS

- ⚡ **7+ years** of professional experience including 4+ years in AI/ML engineering
- ⚡ Achieved **Gold Badges** for Python, Problem Solving, and Statistics on HackerRank
- ⚡ Secured **Rank 69 out of 7,573 teams** in Kaggle ML Challenge
- ⚡ Led teams of **8–15 members** across enterprise AI projects
- ⚡ Delivered **50% model performance improvements** in production systems (BERT, RoBERTa)

- ⚡ Built systems handling **4,500+ daily API requests**
- ⚡ Processed **100K+ daily records** via Airflow pipelines
- ⚡ Achieved **38% accuracy boost** in RAG systems
- ⚡ Reduced training time by **35%** using Optuna optimization
- ⚡ Resolved **28 critical production issues** across Fortune 500 client engagements